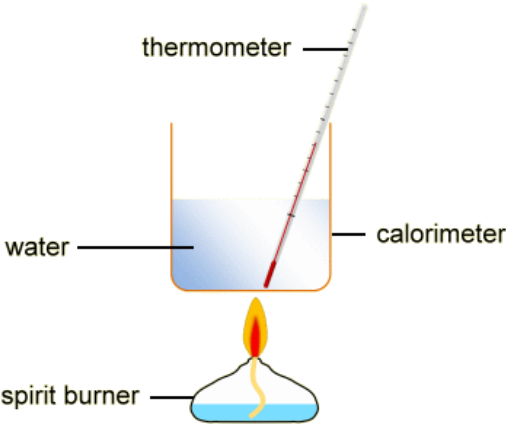


Section B

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7	(a)	(i)		moles CaO = $\frac{1.90}{56}$ = 0.0339 (1) moles HCl = $\frac{50 \times 1.40}{1000}$ = 0.070 (1) 0.07 mol HCl would neutralise 0.035 mol CaO so acid in excess (1)		3		3	1 1	3
		(ii)		4284.5		1		1		
		(iii)		$\frac{mc\Delta T}{n} = \frac{4284.5}{0.0339}$ (1) -126.4 kJ mol ⁻¹ both sign and value needed (1)		2		2	1	
		(iv)		Hess diagram shown with arrows in correct direction (1) ignore products of reactions $\Delta_r H = 126.4 - 196 = -69.6$ kJ mol ⁻¹ (1) <i>(using value given in question 110 - 196 = -86 kJ mol⁻¹)</i>		2		2	1	
		(v)		award (1) each for any two of following • suitable apparatus to minimise heat losses e.g. lid/ polystyrene container • thermometer reading to 0.1°C / graduations to allow reading to less than 0.5°C • use a burette since it can be read to 0.05 cm ³			2	2		2

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		$\text{CH}_3\text{OH(l)} + 1\frac{1}{2}\text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(l)}$ ignore state symbols, do not allow multiples		1		1		
		(ii)		liquid in burner with flame (1) thermometer in water and suitable container (1) 	2			2		2

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)		bonds broken $(\text{C}=\text{C}) + 4(\text{C}-\text{H}) + 3(\text{O}=\text{O}) = 614 + 4(\text{C}-\text{H}) + 1485$ (1) bonds made $4(\text{C}=\text{O}) + 4(\text{O}-\text{H}) = 3196 + 1860 = 5056$ (1) $2099 + 4(\text{C}-\text{H}) - 5056 = -1387$ (1) average C—H bond enthalpy = $\frac{1570}{4} = 392.5 / 393 \text{ kJ mol}^{-1}$ (1)		4		4	3	
		(ii)		ethene is a gas / not a liquid			1	1		1
				Question 7 total	2	13	3	18	7	8